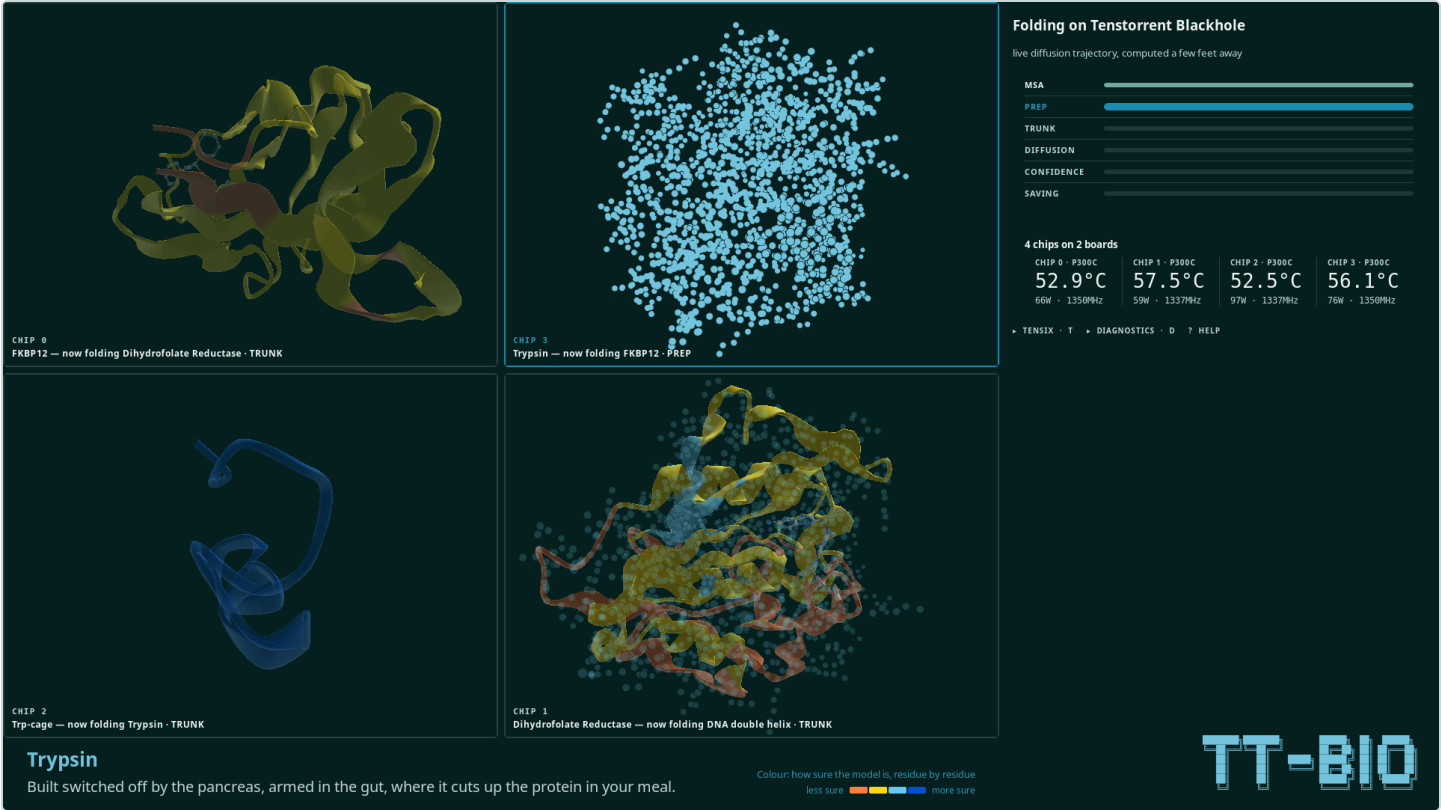


A protein is a chain of amino acids that only works once it has folded into a specific three-dimensional shape. Predicting that shape from the sequence alone was an open problem in biology for fifty years. This booth does it in front of you — [on the four Blackhole chips a few feet away](#), in seconds, with nothing pre-recorded.



**The real application, on real silicon.** Four Blackhole chips (two p300c boards), each folding a different molecule at its own point in the pipeline. Structures are drawn as a cartoon — helices as ribbons, sheets as arrows. The loose dots are the model's actual denoising trajectory, streamed one step at a time as it condenses into structure.

WHAT YOU ARE WATCHING

The model starts from **pure noise** — a cloud of atoms in random positions — and removes a little of that noise at a time, guided by what it has learned about how real proteins are built. Every dot on screen is a genuine intermediate step, sent from the chip as it happens.

Nothing here is an animation of a finished answer. When a fold has no coordinates yet, the booth holds the **previous real structure**, dimmed and labeled, rather than inventing something to show you.

COLOR = THE MODEL'S CONFIDENCE



Each residue is colored by **pLDDT** — the model scoring its own certainty. Blue regions it is sure about; orange it is guessing. An honest model tells you where it is weak, and this one is shown doing exactly that.

THE SIX STAGES ON THE PANEL

|            |                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MSA        | related sequences from other organisms — skipped here, which is why some targets come back orange                                                                  |
| PREP       | builds the model's input features                                                                                                                                  |
| TRUNK      | how every residue relates to every other. Makes <b>no coordinates</b> — so a big protein shows nothing new at first, and the booth keeps the previous structure up |
| DIFFUSION  | the part you watch: noise becomes structure                                                                                                                        |
| CONFIDENCE | the model scores itself — where the colors come from                                                                                                               |
| SAVING     | writes the result out as mmCIF                                                                                                                                     |

**The science is not ours.** This booth exists to show other people's work running on Tenstorrent hardware. Folding is **tt-bio** (MIT) by Moritz Thüning; the demo around it is Apache-2.0. Structures are the model's own output, written as mmCIF and drawn here — no reference structure is ever shown in its place.

WHAT IT FOLDS

| MOLECULE              | SIZE    | WARM   |
|-----------------------|---------|--------|
| Trp-cage              | 20 res  | 4.6 s  |
| DNA double helix      | —       | 4.8 s  |
| Transfer RNA          | —       | 6.9 s  |
| FKBP12 + binder       | 107 res | 9.7 s  |
| DHFR + methotrexate   | 187 res | 14.5 s |
| Trypsin + benzamidine | 223 res | 17.4 s |
| Albumin the long one  | 585 res | 95.5 s |

Measured on this hardware, warm. Three fold a **bound ligand** alongside the protein — including DHFR with methotrexate, a chemotherapy drug, shown sitting in the pocket it blocks. Albumin is the long one: at 585 residues it is what a whole, real, drug-carrying protein costs.

Four chips, four proteins — not one protein four times faster.

A single fold runs on a single chip; that is tt-bio's documented limit, not something this demo works around. Four chips buy four folds at once, and a shorter wait for whatever you pick.

STARTING IT

```
# the normal way — double-click
# "tt-bio Protein Folding Demo"

# or, from a terminal:
/opt/tt-bio-demo/scripts/run-demo.sh
/opt/tt-bio-demo/scripts/run-demo.sh --solo
```

A booth with more than one chip **comes up in the 2x2 grid**; one chip comes up showing a single large protein. `--solo` forces the single view either way, `--quad` forces the grid, and **Q** toggles at runtime. The launcher starts **both** halves — the compute daemon and the UI — and tears the daemon down when it exits.

STOPPING IT

**Ctrl** + **Q**, or **Ctrl** + **C** in its terminal. One INFO line about a `tt-smi` sample killed by a signal is **normal** on a clean stop.

Signaling by pid does nothing — the launcher's trap cannot run until the UI returns. Signal the whole process group:

```
kill -INT -$(ps -o ppid= -p <pid> | tr -d ' ')
```

KEYS

|                                    |                                                                            |
|------------------------------------|----------------------------------------------------------------------------|
| <b>?</b> or <b>F1</b>              | the help card, from any screen                                             |
| <b>Q</b>                           | quad view — all four chips at once                                         |
| <b>T</b>                           | Tensix activity — live core grids                                          |
| <b>D</b>                           | diagnostics — the live protocol log                                        |
| <b>Esc</b>                         | close the card, or either rail panel                                       |
| <b>Ctrl</b> + <b>G</b>             | easter egg: the Tenstorrent mark, computed on a chip                       |
| <b>Ctrl</b> + <b>F</b>             | leave / return to fullscreen                                               |
| <b>Ctrl</b> + <b>A</b>             | restart the booth with affinity Q&A enabled (one-way; no-op if already on) |
| <b>Ctrl</b> + <b>Q</b>             | quit                                                                       |
| any other key<br>or a tap anywhere | wake the booth and show what it folds                                      |

RUNNING IT UNATTENDED

For a booth that must survive all day, let systemd own the compute daemon so it comes back by itself:

```
systemctl --user enable --now tt-bio-demo
systemctl --user status tt-bio-demo
systemctl --user restart tt-bio-demo
```

It restarts **on failure only** — if you stop it deliberately it stays stopped. The UI is still launched separately, from the desktop entry.

BEFORE DOORS OPEN

- ☐ **Four chips present.** `tt-smi -s` lists four entries across two `board_id` s. Three means one did not come up.
- ☐ **Fold each target once.** This is the one that gets skipped. It warms the kernel cache so no visitor watches a cold compile.
- ☐ **Help card opens.** Press **?**. It is what a visitor reads when you are talking to someone else.

THE RIGHT-HAND RAIL

- Pipeline**  
One row per stage — msa, prep, trunk, diffusion, confidence, saving. The bright row is running now; diffusion owns most of the bar. Clears itself after 20 s of silence, so a dead daemon cannot leave a bar frozen all day.
- Chips**  
Temperature, power and clock per chip, from a `tt-smi` snapshot every two seconds — sampled on the UI's own thread, never through the daemon, so a wedged daemon still leaves the hardware readout live.
- Tensix activity — T**  
One animated core grid per chip. Only the chip actually running the fold animates the work, and the header names it.
- Diagnostics — D**  
Every protocol event as it arrives, with a plain-language line for each stage. Closes itself after five minutes of no input.
- Help card — ?**  
Works from any screen without pausing the fold. Closes after a minute.

WHEN A VISITOR TAPS A PROTEIN

It gets folded **next** — never "now". The tap goes to the head of the queue and lands on the next chip to come free. Nothing already running is canceled, because tearing a fold down mid-device-operation is a known instability source.

Measured: **0.25–5.5 s** with all four chips busy (median ≈ 2 s). And most taps queue nothing at all — with six targets on four chips the protein is usually already folding, so that cell simply takes focus. The booth acknowledges every tap immediately on a one-line notice.

**Never run `tt-smi -r` on a shared machine.** A chip missing from telemetry may have been deliberately quarantined by the daemon for temperature. Check `tt-smi -s` first.

**Do the first fold before doors open — about 90 s per target.** Every time on the front of this sheet is measured **warm**. On a machine that has never folded that target, the kernels compile first: measured **94.5 s** against 9.4 s warm. Fold each one you plan to show, once, while nobody is watching.

QUICK FIXES

|                                           |                                                                                                      |
|-------------------------------------------|------------------------------------------------------------------------------------------------------|
| A cell says <b>TRUNK</b> and looks frozen | Working as designed — trunk makes no coordinates. It is holding the last real structure and says so. |
| A chip missing from the chips panel       | Check <code>tt-smi -s</code> . It may be quarantined for temperature on purpose.                     |
| Panels frozen, protein still moving       | Telemetry is sampled separately from the daemon. Restart the daemon; the UI can stay up.             |
| Everything stopped                        | <b>Ctrl</b> + <b>Q</b> , then relaunch from the desktop entry.                                       |

- ☐ **Weights on disk.** The venue is offline — nothing downloads at the booth. `doctor.sh` checks they are present *and* intact; `dpkg -l` only says a package is installed.
- ☐ **Fullscreen.** **Ctrl** + **F**. A multi-chip booth is already in the grid; **Q** toggles.
- ☐ **Leave it alone for a minute.** The attract loop should start folding on its own, unprompted.